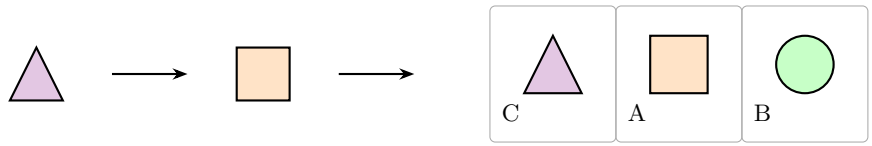
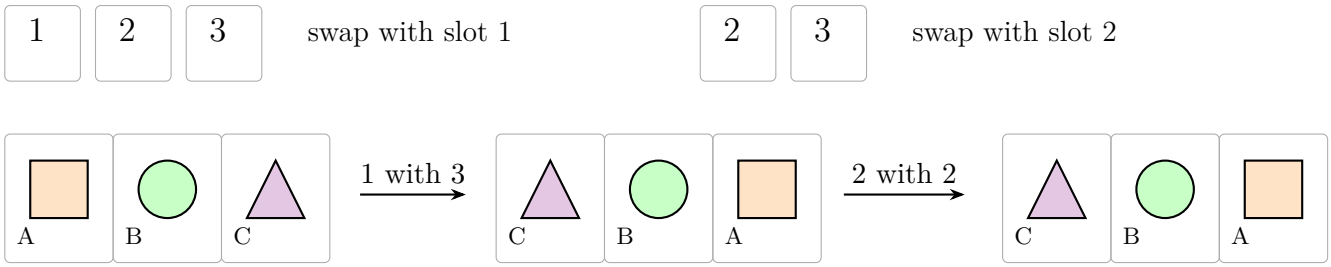
Mix the three picture cards in a cup. Each card has the same chance. Draw the first without looking and keep it out. Draw the second from the two left. Put the last card at the end. Return all cards for a new row.

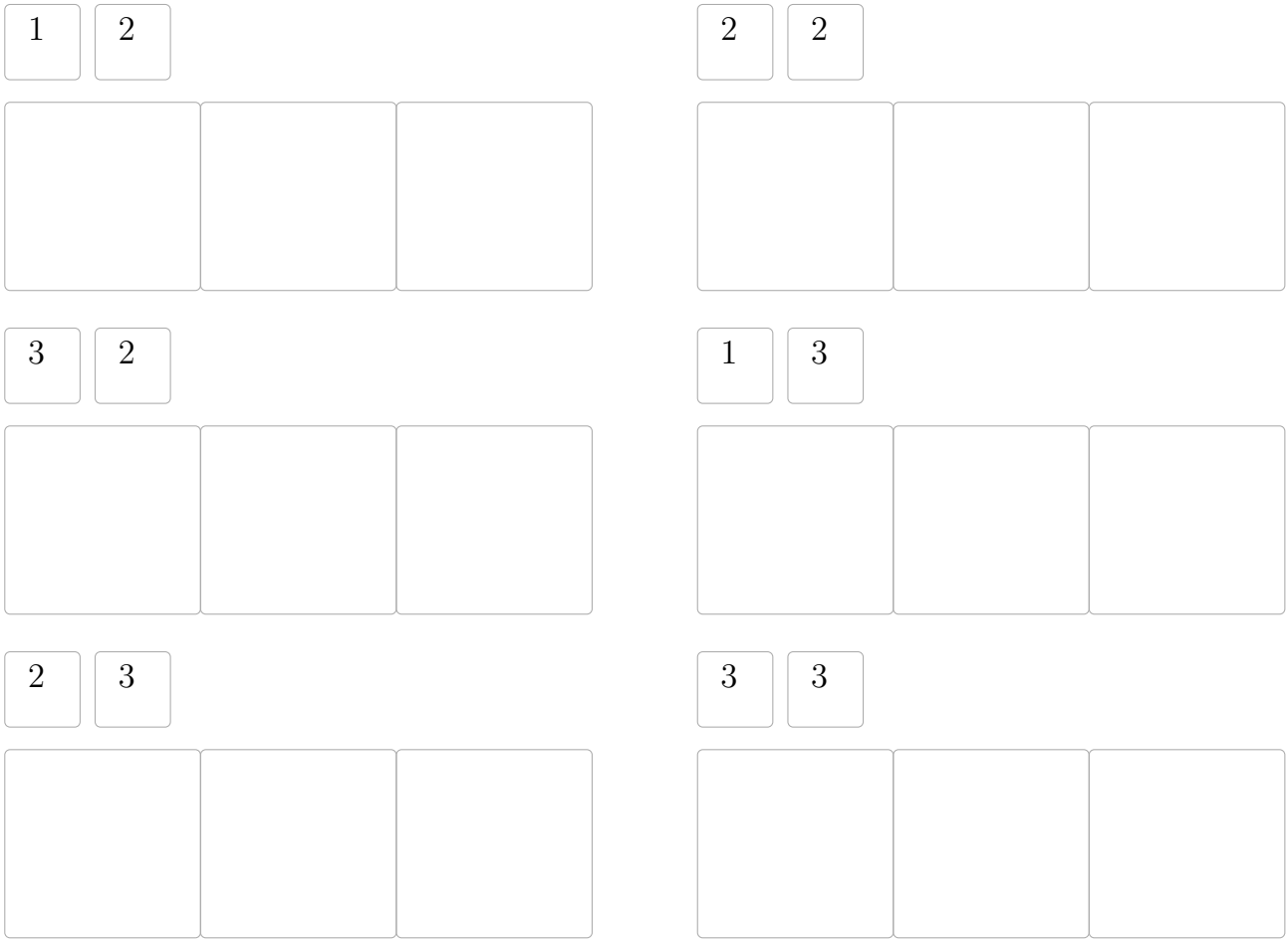


Problem 1: Find every possible final row. Could the chooser give any row more chance than another?

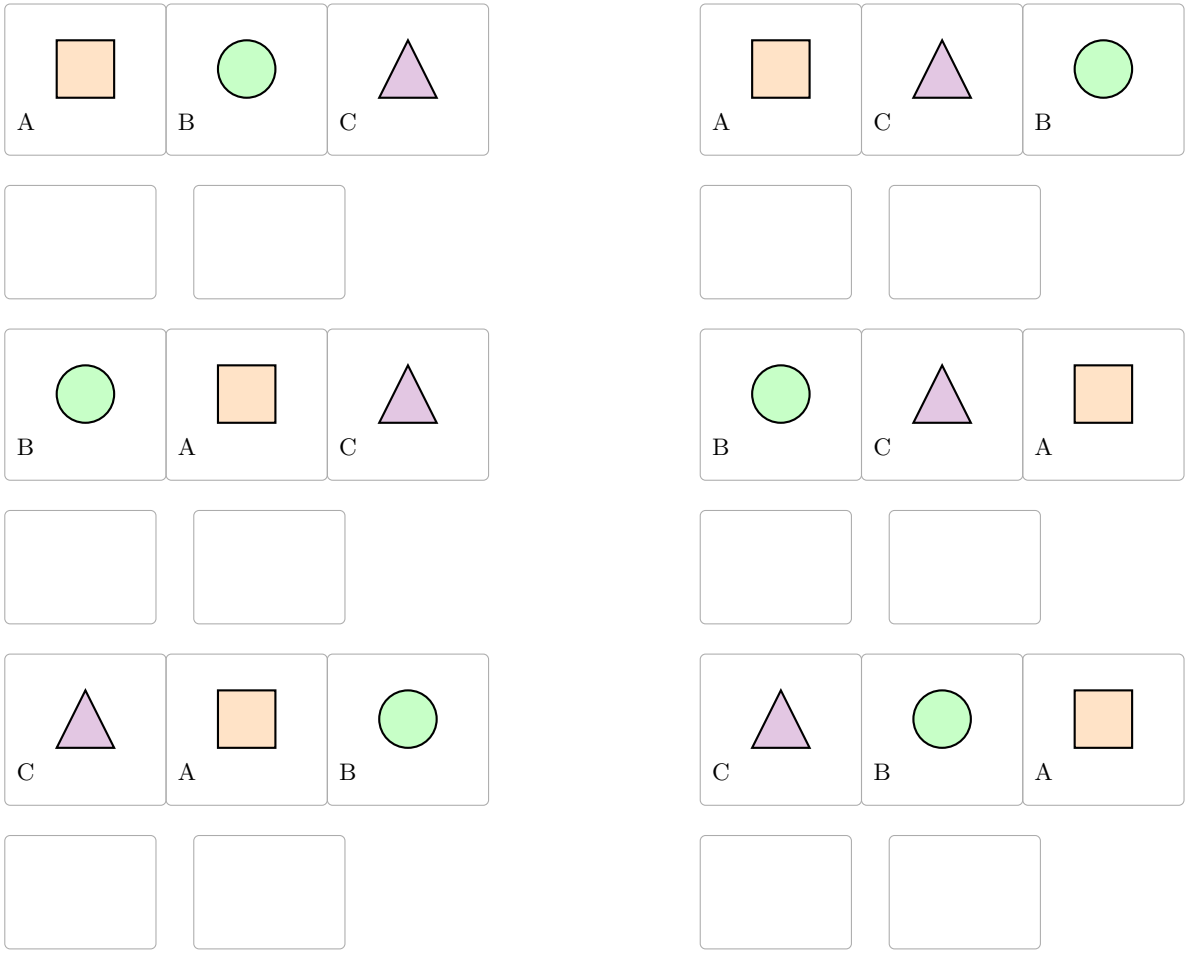
Start A B C. Mix tickets 1, 2, 3 in the cup; draw and swap its slot with slot 1. Empty the cup, then mix tickets 2, 3; draw and swap its slot with slot 2. At each draw, all tickets in the cup have equal chances. Reset both sets for every new row. A slot may swap with itself.



Problem 2: Start with A B C. Match each two-ticket story to the final row it makes. Does this give each row the same chance?



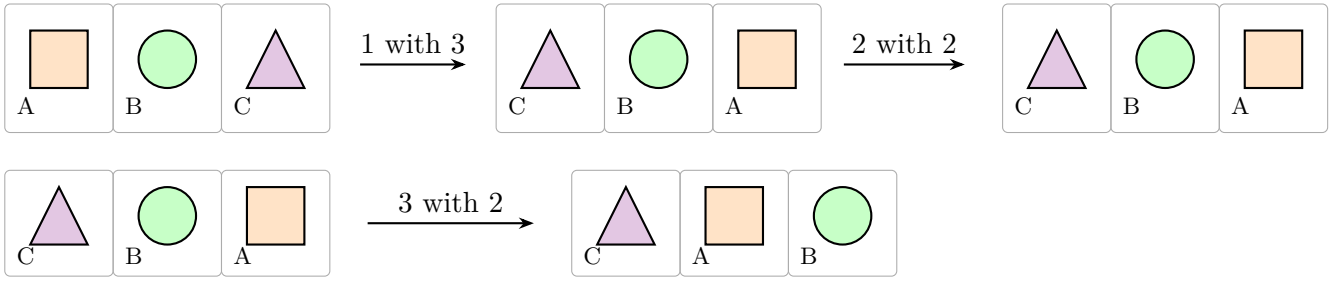
Problem 3: Start with B A C and use the same two-swap rule. Can you still make every row? Find the ticket story for each target.



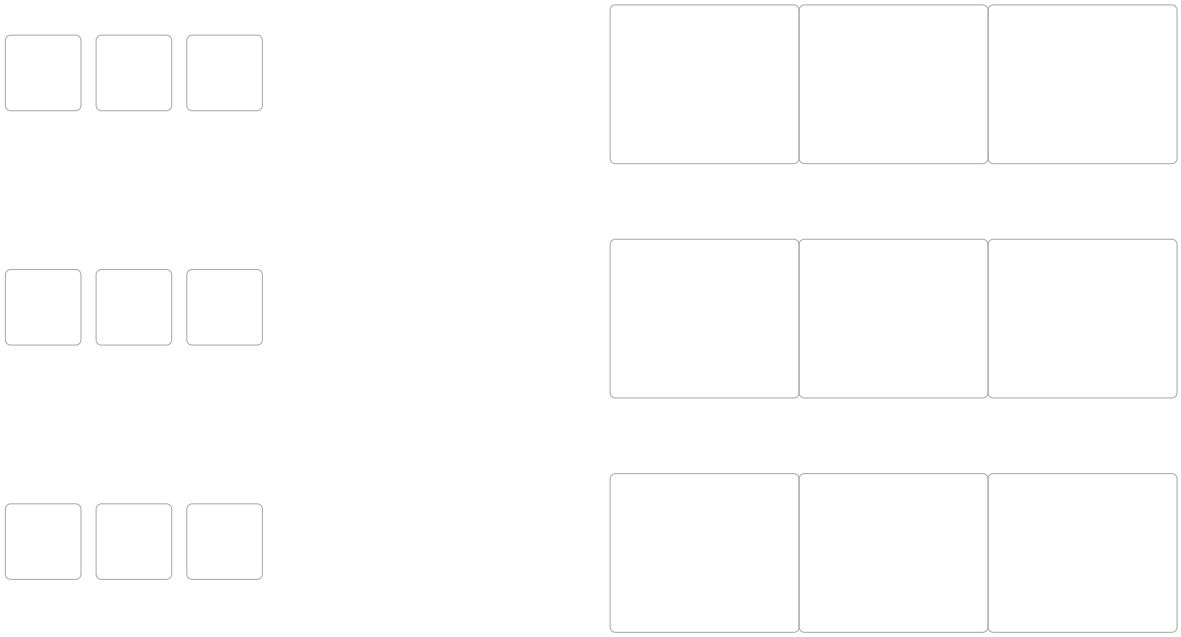
Problem 4: Shuffle six times with the two-swap rule. A friend says six shuffles must show all six rows once. Is the friend right?



Change the rule: use tickets 1, 2, 3 at every swap. Swap slot 1, then slot 2, then slot 3 with the chosen slot. Replace the ticket and mix after each draw.



Problem 5: Start with A B C each time. Find several different three-ticket stories that give the same final row.



Problem 6: There are 27 equally likely three-ticket stories. Could this rule give all six rows the same chance?